DBMS Lab 09 - Task 2: Key-Value Database (Redis)

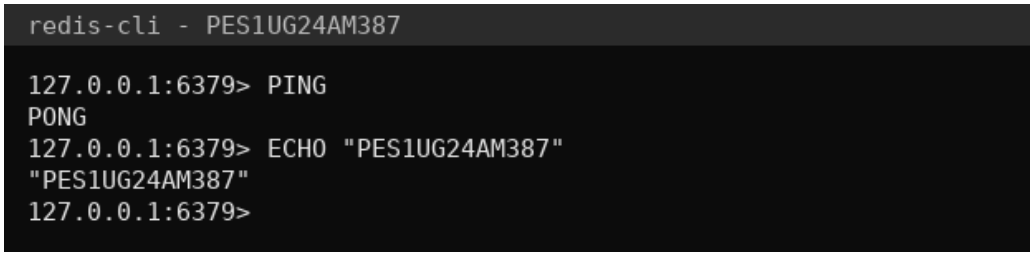
PromptPal - An AI Chat Platform

SRN: PES1UG24AM387

Each question was run from an empty database (FLUSHDB, DBSIZE = 0) in redis-cli; ECHO prints the SRN at the start of every screenshot. Complete command list: PES1UG24AM387_RedisLab.txt.

Session start

```
PING
ECHO "PES1UG24AM387"
```

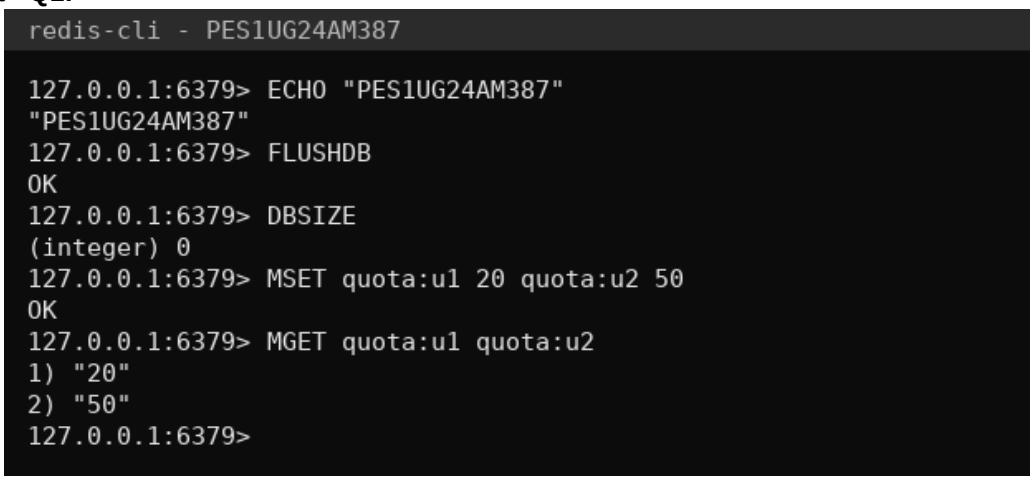


Q1. Daily quota (Easy)

Commands:

```
ECHO "PES1UG24AM387"
FLUSHDB
DBSIZE
MSET quota:u1 20 quota:u2 50
MGET quota:u1 quota:u2
```

Screenshot - Q1:



Replies explained:

• MSET stores both allowances in one command (reply OK).

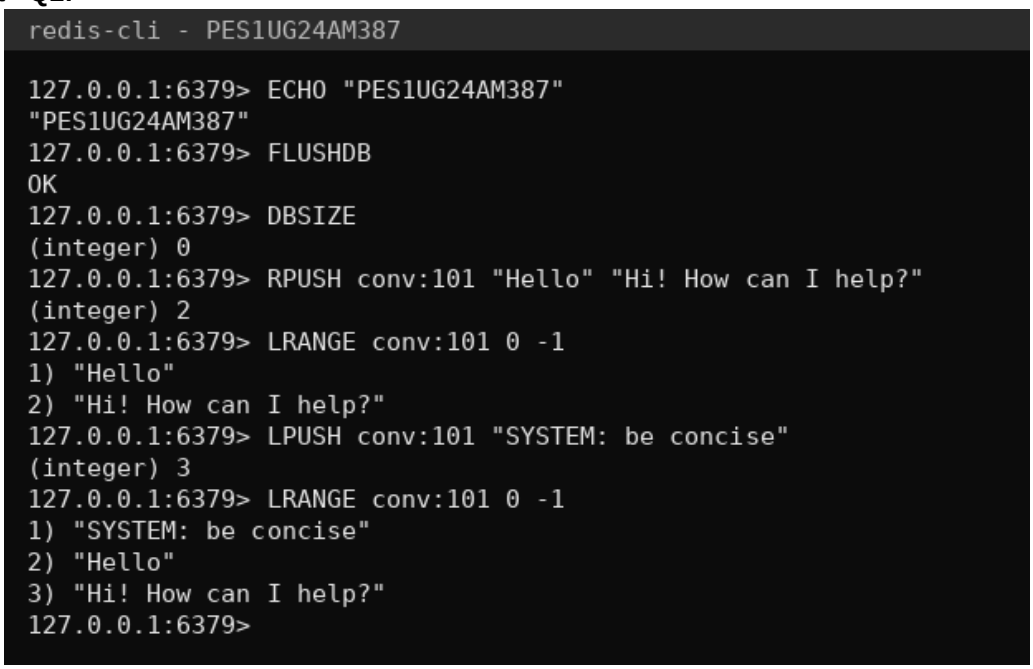
• MGET returns them in the order asked: 1) "20" for quota:u1, 2) "50" for quota:u2.

Q2. Start a chat (Easy)

Commands:

```
ECHO "PES1UG24AM387"
FLUSHDB
DBSIZE
RPUSH conv:101 "Hello" "Hi! How can I help?"
LRANGE conv:101 0 -1
LPUSH conv:101 "SYSTEM: be concise"
LRANGE conv:101 0 -1
```

Screenshot - Q2:



Replies explained:

• RPUSH appends both messages to the end in the order given, so "Hello" is first (reply 2 = new length).

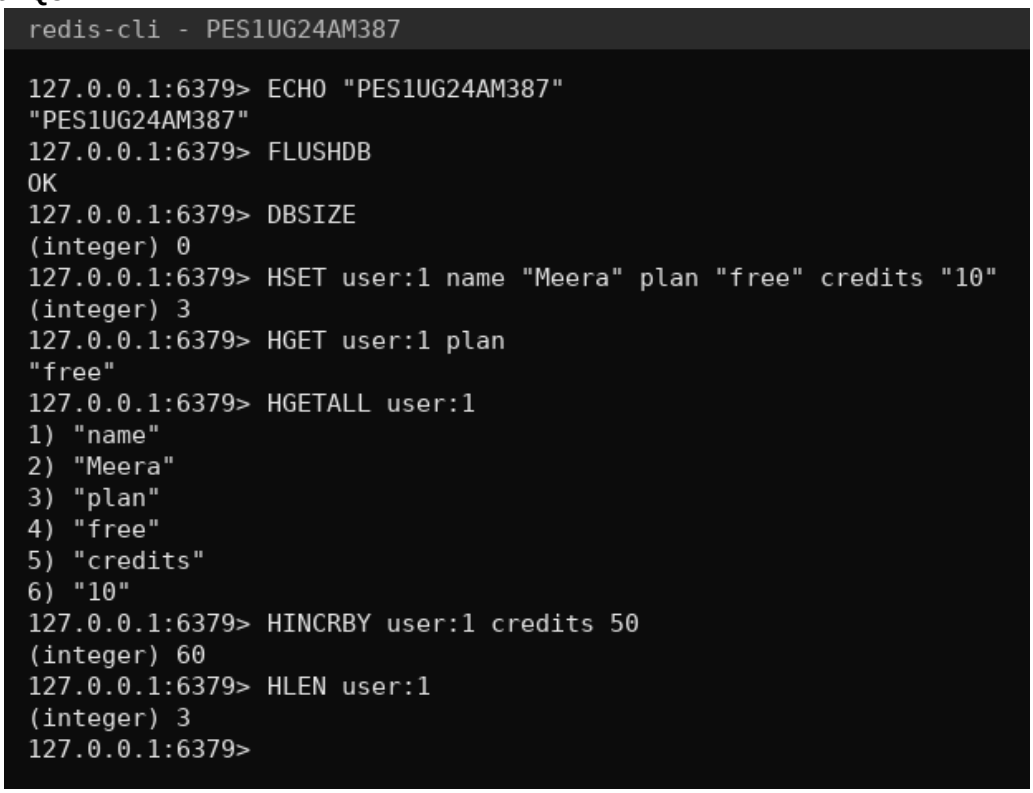
• LPUSH places "SYSTEM: be concise" at the head without changing the order of the existing messages.

Q3. User profile (Medium)

Commands:

```
ECHO "PES1UG24AM387"
FLUSHDB
DBSIZE
HSET user:1 name "Meera" plan "free" credits "10"
HGET user:1 plan
HGETALL user:1
HINCRBY user:1 credits 50
HLEN user:1
```

Screenshot - Q3:



Replies explained:

• HSET replies (integer) 3: three new fields were created.

• HINCRBY credits 50 performs the addition in Redis and replies 60, the new balance.

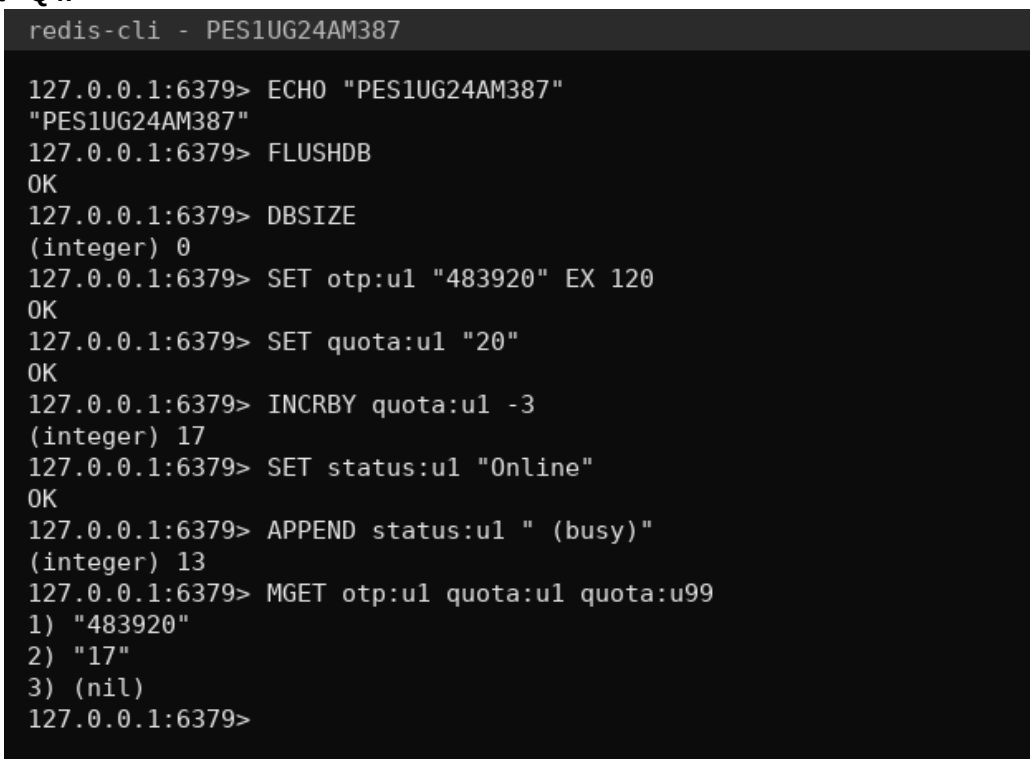
• HLEN replies 3: the profile has three fields.

Q4. OTP and quota (Medium)

Commands:

```
ECHO "PES1UG24AM387"
FLUSHDB
DBSIZE
SET otp:u1 "483920" EX 120
SET quota:u1 "20"
INCRBY quota:u1 -3
SET status:u1 "Online"
APPEND status:u1 " (busy)"
MGET otp:u1 quota:u1 quota:u99
```

Screenshot - Q4:



Replies explained:

• SET ... EX 120 stores the OTP and its 120-second expiry in one command.

• INCRBY quota:u1 -3 subtracts inside Redis and replies 17.

• APPEND replies (integer) 13: the new length in characters of status:u1, now "Online (busy)".

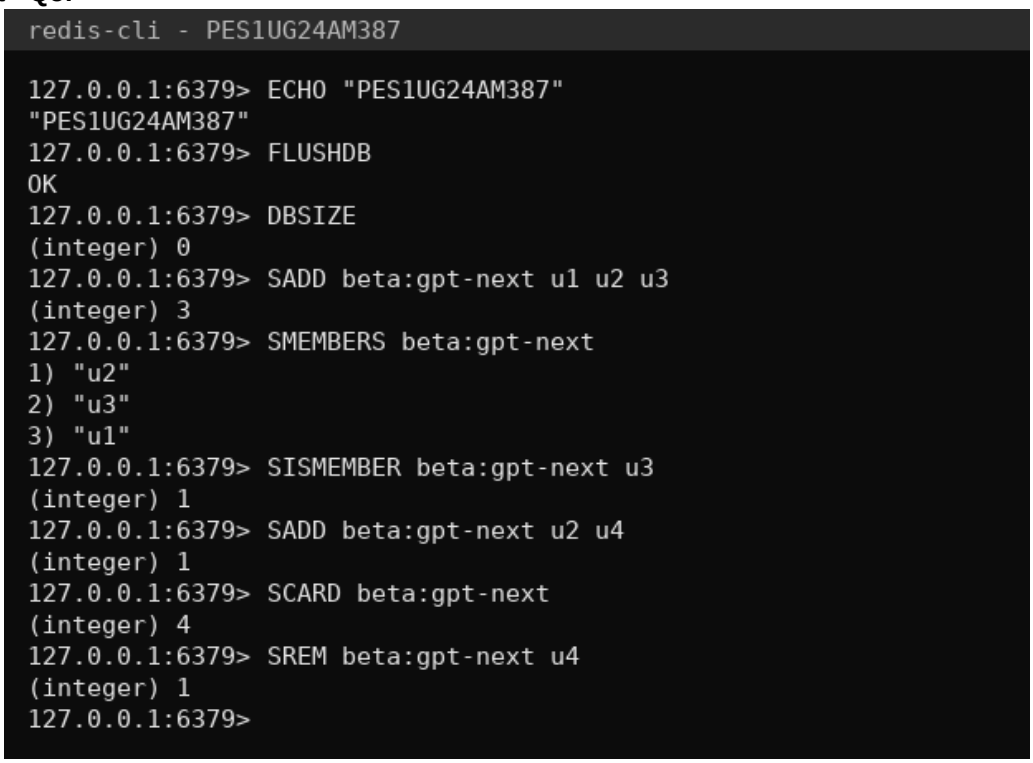
• MGET shows "483920", "17" and (nil) for quota:u99, because that key does not exist.

Q5. Beta program (Medium)

Commands:

```
ECHO "PES1UG24AM387"
FLUSHDB
DBSIZE
SADD beta:gpt-next u1 u2 u3
SMEMBERS beta:gpt-next
SISMEMBER beta:gpt-next u3
SADD beta:gpt-next u2 u4
SCARD beta:gpt-next
SREM beta:gpt-next u4
```

Screenshot - Q5:



Replies explained:

• SADD u2 u4 replies (integer) 1: only u4 is new; u2 was already a member and a set holds each member once.

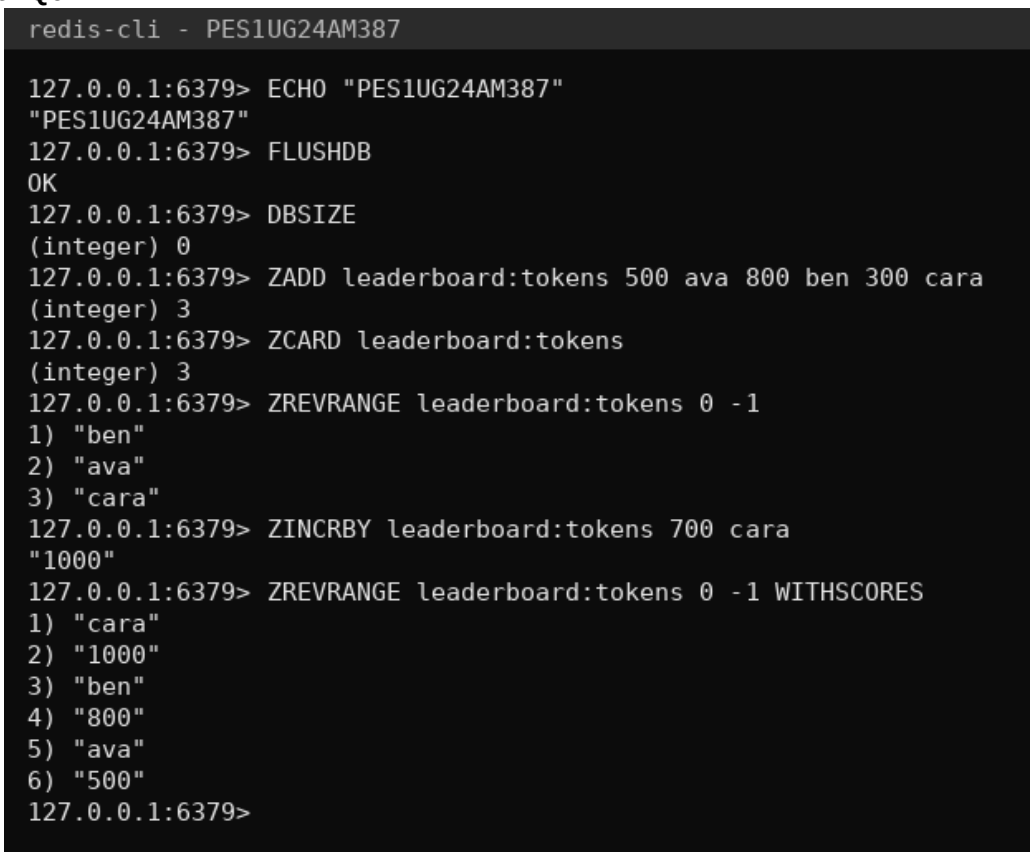
• SCARD replies 4 (u1, u2, u3, u4); SREM u4 replies 1 (one member removed).

Q6. Leaderboard (Medium)

Commands:

```
ECHO "PES1UG24AM387"
FLUSHDB
DBSIZE
ZADD leaderboard:tokens 500 ava 800 ben 300 cara
ZCARD leaderboard:tokens
ZREVRANGE leaderboard:tokens 0 -1
ZINCRBY leaderboard:tokens 700 cara
ZREVRANGE leaderboard:tokens 0 -1 WITHSCORES
```

Screenshot - Q6:



Replies explained:

• ZADD replies 3 (three new members); ZCARD replies 3.

• ZINCRBY 700 cara replies "1000" (300 + 700).

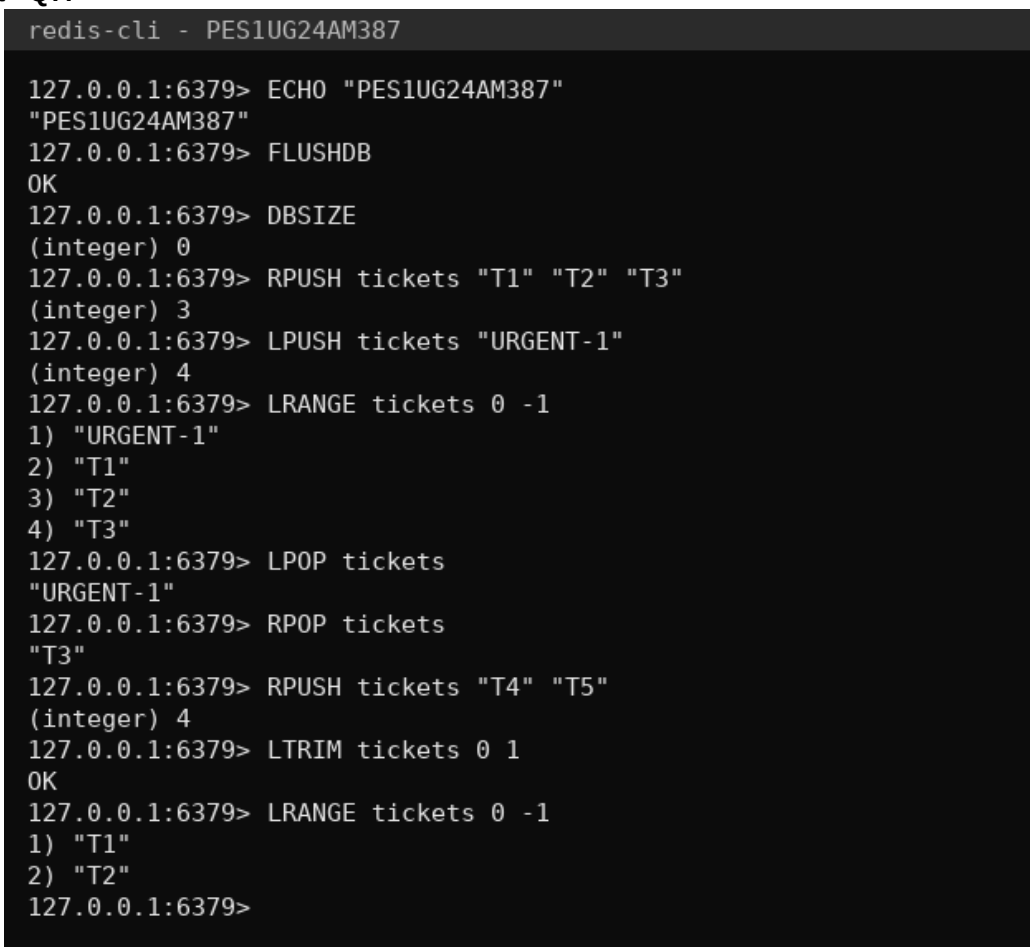
• Cara is #1 now with 1000 tokens, ahead of ben (800) and ava (500).

Q7. Support ticket queue (Hard)

Commands:

```
ECHO "PES1UG24AM387"
FLUSHDB
DBSIZE
RPUSH tickets "T1" "T2" "T3"
LPUSH tickets "URGENT-1"
LRANGE tickets 0 -1
LPOP tickets
RPOP tickets
RPUSH tickets "T4" "T5"
LTRIM tickets 0 1
LRANGE tickets 0 -1
```

Screenshot - Q7:



Replies explained:

• After RPUSH T1 T2 T3 and LPUSH URGENT-1 the queue is URGENT-1, T1, T2, T3.

• LPOP returns "URGENT-1" (the front ticket); RPOP returns "T3" (the back ticket).

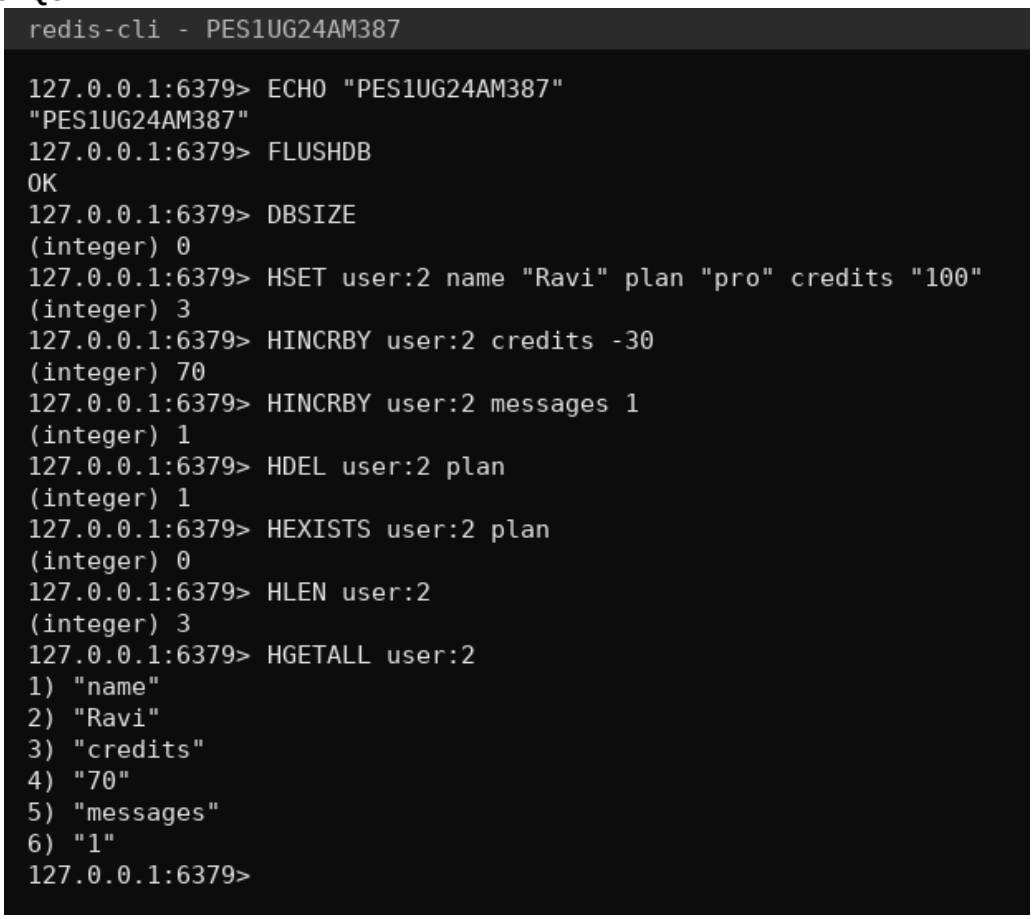
• After RPUSH T4 T5 the queue is T1, T2, T4, T5; LTRIM 0 1 keeps T1 and T2.

Q8. Updating a profile (Hard)

Commands:

```
ECHO "PES1UG24AM387"
FLUSHDB
DBSIZE
HSET user:2 name "Ravi" plan "pro" credits "100"
HINCRBY user:2 credits -30
HINCRBY user:2 messages 1
HDEL user:2 plan
HEXISTS user:2 plan
HLEN user:2
HGETALL user:2
```

Screenshot - Q8:



Replies explained:

• HINCRBY credits -30 replies 70; HINCRBY messages 1 creates the missing field from 0 and replies 1.

• HDEL plan replies 1; HEXISTS plan then replies 0 (absent).

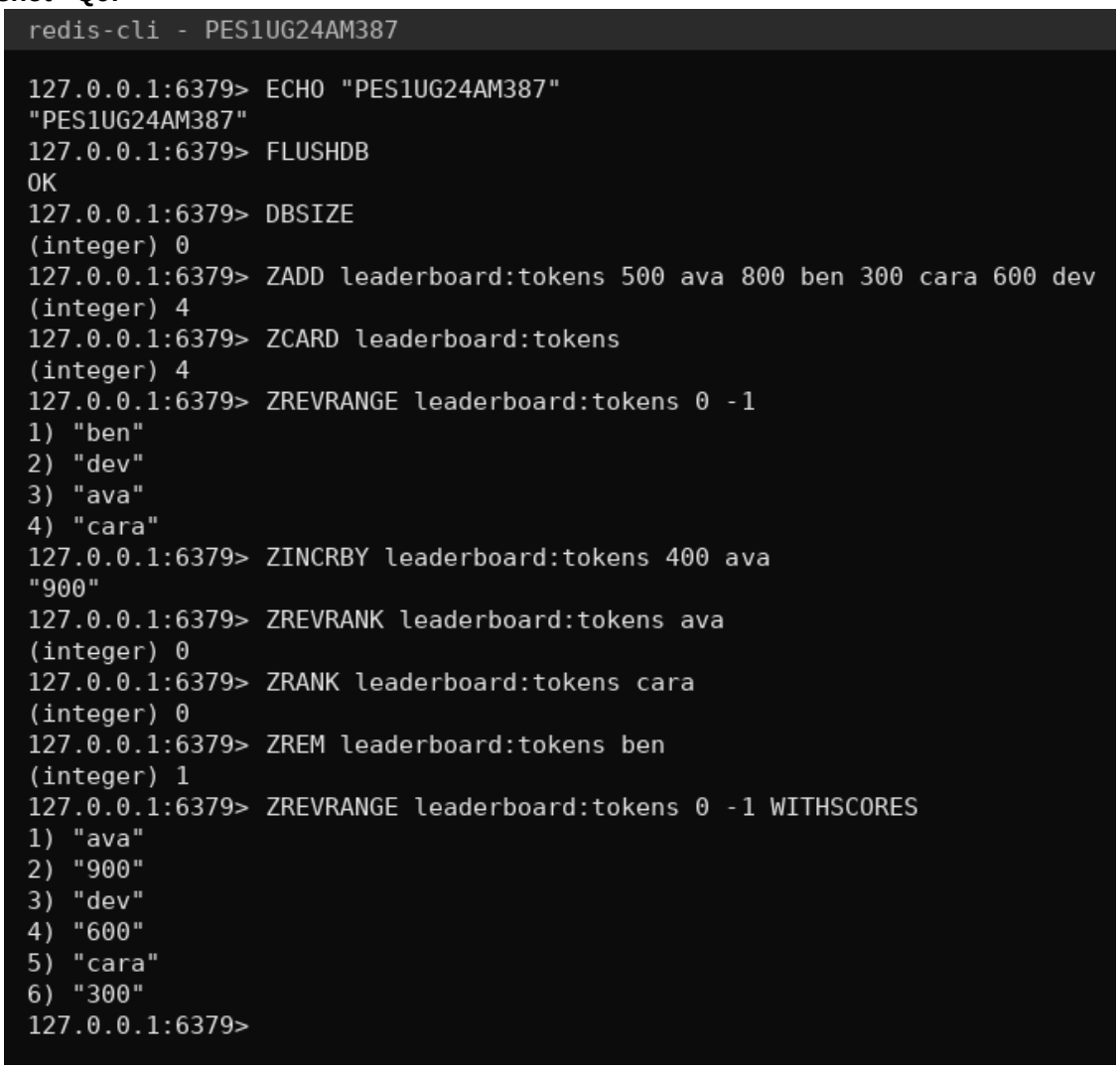
• HLEN replies 3: name, credits and messages remain.

Q9. Leaderboard race (Hard)

Commands:

```
ECHO "PES1UG24AM387"
FLUSHDB
DBSIZE
ZADD leaderboard:tokens 500 ava 800 ben 300 cara 600 dev
ZCARD leaderboard:tokens
ZREVRANGE leaderboard:tokens 0 -1
ZINCRBY leaderboard:tokens 400 ava
ZREVRANK leaderboard:tokens ava
ZRANK leaderboard:tokens cara
ZREM leaderboard:tokens ben
ZREVRANGE leaderboard:tokens 0 -1 WITHSCORES
```

Screenshot - Q9:



Replies explained:

• ZADD replies 4; initial order (highest first): ben, dev, ava, cara.

• ZINCRBY 400 ava replies "900", so ava becomes the top player.

• ZREVRANK ava replies 0 (top); ZRANK cara replies 0 (lowest score, 300).

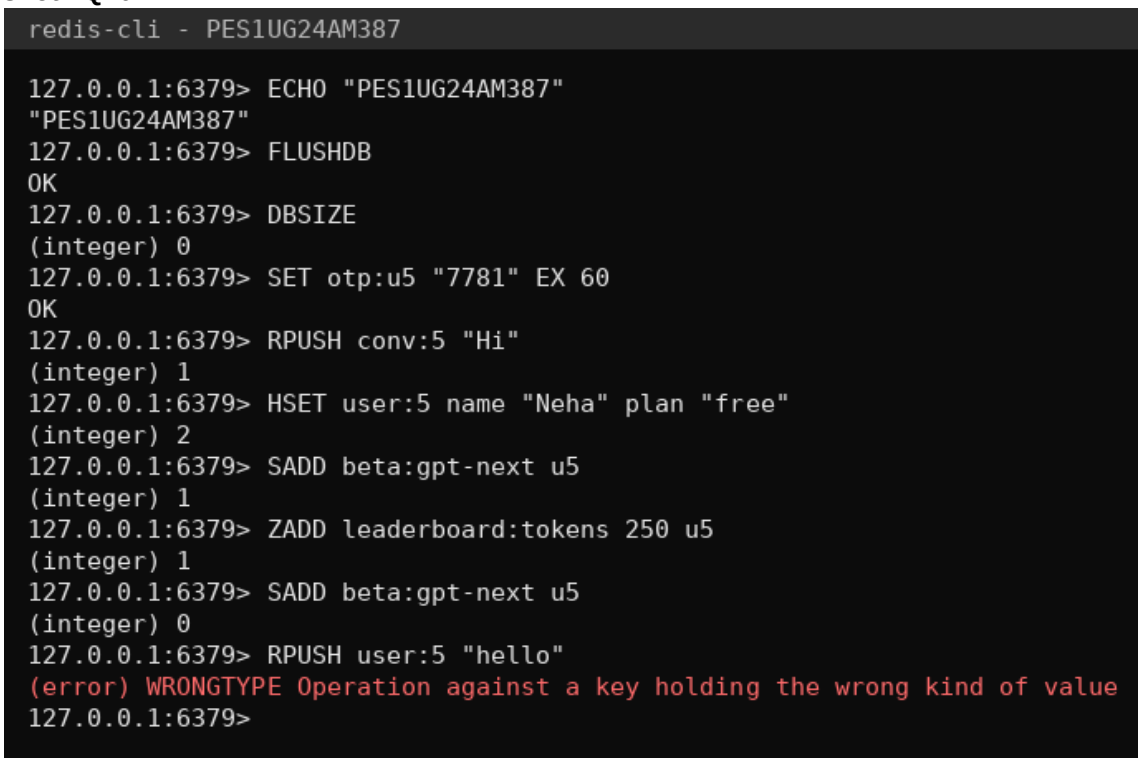
• After ZREM ben: ava 900, dev 600, cara 300.

Q10. Pick the right type (Hard)

Commands:

```
ECHO "PES1UG24AM387"
FLUSHDB
DBSIZE
# 1. String (single value with expiry)
SET otp:u5 "7781" EX 60
# 2. List (keeps arrival order; new messages go at the end)
RPUSH conv:5 "Hi"
# 3. Hash (named fields under one key)
HSET user:5 name "Neha" plan "free"
# 4. Set (unique members, no order)
SADD beta:gpt-next u5
# 5. Sorted Set (members ordered by numeric score)
ZADD leaderboard:tokens 250 u5
# 6. Repeat part 4
SADD beta:gpt-next u5
# 7. List operation on a Hash key
RPUSH user:5 "hello"
```

Screenshot - Q10:



Replies explained:

• 1. String - SET otp:u5 "7781" EX 60 (one value that expires).

• 2. List - RPUSH conv:5 "Hi" (keeps arrival order; later messages go after it).

• 3. Hash - HSET user:5 name "Neha" plan "free" (named fields under one key).

• 4. Set - SADD beta:gpt-next u5 (unique members, no order).

• 5. Sorted Set - ZADD leaderboard:tokens 250 u5 (members ordered by score).

• 6. Running SADD beta:gpt-next u5 again returns (integer) 0: u5 is already a member, and a set never stores a duplicate, so nothing new was added. This is not an error.

• 7. RPUSH user:5 "hello" fails with WRONGTYPE Operation against a key holding the wrong kind of value: user:5 is a Hash, and a key can hold only one type, so List commands cannot be used on it.